MIL-R-52169B(EL)
4 November 1975
SUPERSEDING
MIL-R-52169A(EL)
27 August 1971

MILITARY SPECIFICATION

REELING MACHINE, CABLE, ENGINE DRIVEN RL-207( )/G

This specification is approved for use by all Departments and Agencies of the DOD.

1. SCOPE

1.1 Scope.- This specification covers Reeling Machine, Cable, Engine Driven, RL-207( )/G. (See 6.1 & 6.3)

1.2 Description.- Reeling Machine, Cable, Engine Driven RL-207 ( )/G is a heavy duty reel unit capable of picking up and paying out field cable and field wire at speeds up to 5 miles per hour for pickup and 15 miles per hour for pay out. Power, from a 3 horsepower, 4 cycle, air cooled gasoline engine, is transmitted from the engine to two reel axles by means of belts. Variable reel speed control from zero to maximum rpm is accomplished by slipping belt arrangement. Separate control handles are provided for both the upper and lower reels. The reels axles can accommodate the following reels: two DR-5 reels, one per axle; two DR-15 reels, one per axle; or four RL-159U Reels, two per axle. The reel axles are capable of being operated independently or simultaneous at the discretion of the operator.

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of the issue in effect on date of solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-575 Bolt, Hexagon and Square

FSC 3895

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PPP-B-621	Boxes, Wood, Nailed and Lock Corner.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant.
FF-N-836	Nut, Square, Hexagon, Cap, Slotted Castellated, Clinched, Knurled, and Welded.
FF-W-92	Washer, Metal, Flat (Plain)
QQ-S-781	Strapping, Steel, Flat and Seals

MILITARY

MIL-P-116	Preservation, Packaging Methods of.
MIL-B-117	Bag, Sleeve and Tubing - Interior
MIL-W-6858	Welding: Aluminum, Magnesium, Non-hardening Steels or Alloys, and Titanium, Spot, Seam and Stitch.
MIL-B-7883	Brazing of Steels, Copper Alloys, and Nickel Alloys.
MIL-S-8660	Silicone Compound
MIL-E-10062	Engines, Spare or Installed (Other than Aircraft) Preparation for Shipment and Storage of.
MIL-G-10924	Grease; Automotive and Artillery
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas Metal Arc and Gas Tungsten Arc Aluminum Alloys Readily Weldable excluding Armor
MIL-P-11268	Parts, Materials and Processes used in Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods

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FEDERAL

FED-STD-595 Colors

DRAWINGS

ELECTRONICS COMMAND

SC-DL-322574 Reeling Machine, Cable, Engine Driven.

3. REQUIREMENTS

3.1 First Article of production.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish two First Articles for approval (See 4.3).

3.2 Construction.- The equipment shall be constructed in accordance with the requirements of this specification, and of Drawing and Data List SC-DL-322574 and the requirements of MIL-P-11268 (See 4.15).

3.2.1 Engine.- The power supply for the reeling machine shall be a three net continuous horsepower, air-cooled, 2 cylinder, 4 cycle, Military Standard Engine, Model 2A016-III.

3.2.2 Belts.- Power shall be transmitted to the reel axles by means of a timing belt and double V belts.

3.2.3 Control Levers.- Control levers shall control each reel axle independently. The performance shall be as specified in paragraphs 3.13 thru 3.15.

3.2.4 Hub locks.- When hub locks are engaged the shafts shall not rotate if power should be applied accidently by means of the control levers. Also, when engaged, the hubs shall be aligned so that the loaded reel axles can be easily inserted.

3.3 Cleaning.-

3.3.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable portion of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before parts are assembled. When necessary cleaning shall also be performed after final assembly.

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3.3.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from foreign material.

3.4 Finish, protective.- The reeling machine shall be finished in accordance with applicable requirements of Specifications MIL-F-14072 and MIL-S-8660, and the equipment drawings (See 4.4).

3.5 Marking.- Marking shall conform to Specification MIL-M-13231 (See 4.4).

3.5.1 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipments (See 4.4).

3.5.2 Serial numbers.- Each Reeling Machine, Cable, Engine Driven, RL-207( )/G shall be serial numbered.

3.5.3 Aircraft loading data plate.- The Reeling Machine shall be provided with a data plate in accordance with the drawings, imprinted with the dimensions which locate the center-of-gravity of the machine in the three planes and indicates the gross weight of the machine.

3.6 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements or permanent separation of parts will not occur when the items in which they are used are subjected to shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements.

3.7 Brazing.- Brazing of steels, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883 (See 3.25).

3.8 Tropicalization of material.- Canvas, webbing, wood etc. shall be treated to be resistant to moisture and fungus as indicated on the drawings and as follows.

3.8.1 Treating materials.- Treating materials containing a mercury bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

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3.8.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment during use of the finished items when used for the purpose intended.

3.8.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.8.4 Statement of Treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.

3.9 Welding.- Welding shall conform to MIL-W-6858 and/or MIL-W-45205 and shall be as defined by the drawings.

3.9.1 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved, Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel (See 3.25).

3.10 Weight.- The weight of the equipment, dry, excluding tools, crank, canvas cover, reels and reel axles shall not exceed 480 pounds. (See 4.17)

3.11 Output power.- A minimum of 2 horsepower (hp) shall be available at each of the two reel axles (one at a time) at a continuous axle speed of 151 ± 10 rpm without any slippage of belts and the engine adjusted in accordance with 3.12 and its crankshaft speed adjusted to 2650 ± 75 rpm. (See 4.9.1.1)

3.12 Engine adjustments.- The contractor shall adjust all completed reeling machines as follows:

a. Adjust the engine throttle control for proper operating conditions.

b. Adjust the engine carburetor for the most efficient engine performance.

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3.13 Braking.- With fully loaded reels on the axles and the axles rotating at a speed of 151 + 10 rpm, each control lever, when moved to the braking position shall bring the axles to a complete stop within three (3) seconds. (See 4.9.1.2)

3.14 Neutral position.- With fully loaded reels and the engine operating to deliver a crankshift speed of 2650 ± 75 rpm, there shall be no power transmitted to either axles and no creepage shall occur, when the control levers are in the neutral position and the axles are in the free turning position. (See 4.9.1.3)

3.15 Manual operation.- With the engine not operating and the control levers in the neutral position, the axles shall move freely without binding, catching, or drag when paying out or reeling in wire or cable manually by means of crank GC-4-A. (See 4.9.2)

3.16 Life.- When operated for 300 hours as specified in paragraph 4.9.3, the equipment shall have no physical deterioration which could prevent it from meeting the requirements of 3.11 through 3.15.

3.17 High Temperature.- The equipment shall meet the requirements of Para 3.11, 3.13, 3.14, 3.15, 3.2.4 and 3.25 when tested in accordance with Para 4.13.

3.18 Low Temperature.- The equipment shall meet the requirements of Para 3.11, 3.13, 3.14, 3.15, 3.2.4 and 3.25 when tested in accordance with Para 4.14.

3.19 Humidity.- After being tested as specified in paragraph 4.8, the equipment shall meet all the requirements of 3.11, 3.13, 3.14, 3.15, 3.2.4 and 3.25.

3.20 Rain.- After being tested in accordance with para 4.10 the equipment shall meet the requirements of para 3.11 to 3.15.

3.21 Shock.- The equipment shall have no evidence of physical damage and shall meet all the requirements of 3.11 thru 3.15 and 3.6 after being subjected to the test specified in paragraph 4.11.

3.22 Lubrication.- All bearings, exclusive of the engine and of the brake and control linkage, shall be of the permanently self-lubricated type. All ball bearings, exclusive of the engine, shall be packed with Grease, Automotive and Artillery, GAA, per Specification MIL-G-10924, and shall be sealed-for-life-type. (See 4.16)

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3.23 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically functionally interchangeable, without modification of such items or of the equipment. (See 4.7) Individual items shall not be hand-picked for fit performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.23.1 The canvas cover shall fit the machine without stretching the material. See Para 4.12.

3.24 Technical literature, tools, and running spare parts.- Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.25 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein and shall not contain any of the defects as listed in Para 4.16.

3.2	Construction
3.3	Cleaning
3.6	General requirements for securing of parts
3.9.1	Process for Welding
3.7	Brazing

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection by subsidiary documents. (See 4.4)

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c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.18)

4.3 First Article.- Unless otherwise specified in the contract, the First Article Inspection shall be performed by the contractor.

4.3.1 First Article Units.- The contractor shall furnish two first article units of the complete Reeling Machine, RL-207( )/G.

4.3.2 First Article Inspection.- The first article inspection shall consist of the inspections specified in Table 1 and shall be performed as follows: One of the two first articles shall be subjected to environmental testing in the following order: Humidity, Rain, High Temperature, and Low Temperature. The other first article shall be subjected to performance tests in the following order: shock, life operational, shrinkage of fabric cover. Other testing specified in Table 1 shall be performed on both units and may proceed or follow the above inspections.

TABLE 1 - First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	3.4 & 3.5	4.4
2. Interchangeability	3.23	4.7
3. Humidity	3.19	4.8
4. Rain	3.20	4.10
5. High Temperature	3.17	4.13
6. Low Temperature	3.18	4.14
7. Shock	3.21	4.11
8. Weight	3.10	4.17

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9. Visual & Mechanical	3.25	4.16
10. Braking	3.13	4.9.1.2
11. Neutral Position	3.14	4.9.1.3
12. Hublocks	3.2.4	4.9.1.4
13. Manual	3.15	4.9.2
14. Output Power	3.11	4.9.1.1
15. Life Operational	3.16	4.9.3
16. Shrinkage of Fabric Cover	3.23.1	4.12

4.4 Quality conformance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

ITEM	WHERE REQUIRED
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection.- Equipment shall be assembled in discrete lots for quality conformance inspection, utilizing the procedures on MIL-STD-105 for lot formation. Each lot shall consist of units produced within one month or less. Each unit of each lot of equipment shall be inspected for conformance to all the examinations and test requirements of Table II, Group A Inspection. The quality conformance of each lot shall then be subjected to a verification audit utilizing the procedures of MIL-STD-105 using the S-1 Special Inspection Level, Single sampling plans for normal inspection, and an Acceptable Quality Level (AQL) of 0.5%. Group A Inspection will be performed in any order which is satisfactory to the government, (before performing Group A Inspection, condition equipment as per Para 4.6).

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TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	100% Inspection
Visual Mechanical	3.25	4.16	
Engine Adjustment	3.12	4.6	
Braking	3.13	4.9.1.2	
Neutral Position	3.14	4.9.1.3	
Hublocks	3.2.4	4.9.1.4	
Manual	3.15	4.9.2	

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the single sampling plans for normal inspection procedures of MIL-STD-105. Equipment shall be assembled in discrete lots for quality conformance inspection, utilizing the procedures of MIL-STD-105. Equipment shall be assembled in discrete lots for quality conformance utilizing the procedures of MIL-STD-105 for lot formation. Each lot shall consist of units produced within one month or less. Group B Power Output inspection shall be performed on units that have been subjected to and satisfied the Group A inspection. Group B Interchangeability inspection shall be performed on production lots of parts prior to assembly.

TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL %	Insp Level
Output Power	3.11	4.9.1.1	2.5	II
Interchangeability	3.23	4.7	6.5	S-4

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4.5.3 First Group C Inspection.- Two units shall be selected from the first production lot presented for acceptance and shall be subjected to a complete First Article Inspection per para 4.3 plus the Construction Verification Inspection Para 4.15. This inspection is in addition to the inspection performed on the First Article Units.

4.5.4 Group C Inspection.- This inspection shall consist of the tests specified in Table IV and shall be performed on units that have been subjected to and passed Group A and Group B Inspection requirements. One unit shall be selected for test from the second 100 units produced on contract or fraction thereof and from each succeeding 100 units or fraction thereof.

TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Life	3.16	4.9.?
Shrinkage of fabric cover	3.23.1	4.12
Construction Verification	3.2	4.15

4.5.5 Group C Failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

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4.5.6 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage and wear and replacement of drive belts with new belts.

4.6 Conditioning.- Before performing any operational tests and/or visual mechanical inspection the engine shall be started, adjusted and operated at idling speed for 15 minutes. (See 3.12)

4.7 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.23. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Dimensions of all pulleys and belts.
b. Dimensions of all bearings and shafts.
c. Spacing between reel axle hubs.
d. Dimensions of reel axles.
e. Dimensions of covers and location and sizes of mounting holes; holes for access to ends of drive shafts.
f. Dimensions of hand cranks.
g. Dimensions of brake drums and brake bands.

4.8 Humidity Test.- The equipment shall be tested in accordance with MIL Std 810 Method 507, Procedure II.

4.8.1 Perform the inspections specified in paragraphs 4.9.1, 4.9.1.1, 4.9.1.2, 4.9.1.3, 4.9.1.4, 4.9.2 and 4.16. To insure compliance with Requirements of Para 3.12, 3.13, 3.14, 3.15 and 3.25.

4.8.2 Subject the reeling machine to five continuous 48 hour cycles of the Humidity Test per figure 507-2 of MIL Std 810. No inspections shall be performed during cycling.

4.8.3 Prior to operation of the machine insure that the fiber components within the distributor of the engine are dry and the ignition points are cleaned. Start and adjust the engine for proper performance. Test the reeling machine as specified in Paragraphs 4.9.1, 4.9.1.1, 4.9.1.2, 4.9.1.3, 4.9.1.4, 4.9.2 and 4.4. The machine shall meet the requirements of Para 3.19.

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4.9 Operational.-

4.9.1 Power.- After being conditioned as specified in paragraph 4.6, the equipment shall be tested as follows:

4.9.1.1 Output.- Using a dynamometer or an accurately calibrated electric generator and load board, the equipment shall be tested to determine compliance with the requirements specified in paragraph 3.11. Operation at specified load shall be continuous for not less than one hour for each reel axle.

4.9.1.2 Braking.- With the reel axles operating at a speed of 151 ± 10 rpm, each operating lever shall be moved to the braking position to determine compliance with the requirement specified in paragraph 3.13.

4.9.1.3 Neutral position.- The equipment shall be tested to determine compliance with the requirement specified in paragraph 3.14.

4.9.1.4 Hub locks.- With the hub locks engaged, on each axle and with engine idling a sudden momentary light pressure shall be applied to each operating lever to determine compliance with the requirement specified in paragraph 3.2.4.

4.9.2 Manual operation.- With the hub locks released and the operating lever in the neutral position, both reel axles, loaded, shall be rotated manually one at a time, using the hand crank to determine compliance with paragraph 3.15.

4.9.3 Life.- The equipment shall be tested as follows to determine compliance with paragraph 3.16.

a. The engine shall be conditioned as specified in paragraph 4.6 at the start of each 16 hour period and shall run continuously except for stops required to replenish fuel and to check the level of oil and add oil, if required in the crankcase.

b. The reel axles shall be loaded with a reel (RL-159) containing one mile or more of field wire WD-1/TT on one axle and a drive arrangement for a dynamometer or a generator as described in 4.9.1.1, with a 2 horsepower load applied continuously, on the second axle. After 150 hours of operation the loads shall be alternated with each other on the axles, and the remainder of tests performed with the loads mounted accordingly.

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c. With the engine operating, at 2650 ± 75 rpm, move the operating levers to the operating position, one at a time, starting with the one which controls the axle with the reel installed on it. Each lever shall be held in the fully engaged and non-slipping position. Both axles shall be operated at 151 ± 11 rpm, simultaneously, for a continuous period of 16 hours in each 24 hours (except for steps as noted above). The brakes shall be applied every 60 minutes to stop the rotation of both axles (one at a time) as quickly as possible.

d. Following 300 hours of operation, the equipment shall be disassembled and inspected for excessive wear or damage.

4.10 Rain test.- The equipment, operating, at engine speed of 2650 ± 75 rpm and loaded reels in place shall be tested as follows:

a. Place the equipment, in its normal operating position, in a test chamber maintained at 77° ± 5°F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ± 5°F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for not less than one hour.

b. With the rainfall continuing at the 2 inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for not less than on hour.

c. Rotate the equipment 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for not less than one hour.

d. Continue as in c., until all four sides of the equipment have been exposed to the wind and rain, each side not less than one hour.

e. The equipment shall meet the requirements of 3.20 immediately after the test.

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4.11 Shock.- Raise one end of the equipment to an angle of 45° ± 5°. Drop the unit onto a 4 x 4 inch timber placed to catch the end. The timber shall be backed by concrete or a rigid steel frame. The top of either the steel frame or the concrete shall be at the same level of the unraised end of the equipment. Repeat the drop of each of the remaining 3 sides. At the completion of the test , all parts of the unit shall show no visible deformation and the unit shall operate freely without evidence of shaft misalignment, binding or excessive vibration. Any defects discovered during this test shall be corrected by the contractor on all units on order and additional units tested until the required results are obtained. (See 3.21)

4.12 Test for shrinkage.- The canvas cover shall be inspected for satisfactory fit both before and after the following test: Soak the cover in tap water at normal room temperature for 24 hours and then dry thoroughly at normal room temperature. Place the canvas cover on the equipment and examine for a proper fit. (See 3.23.1)

4.13 High Temperature Test.- The equipment shall be tested in accordance with MIL Std 810, Method 501, Procedure II. Step 6 shall be deleted from the procedure. In step 4 the internal chamber temperature shall be 150°F. In step 7 the operating temperature shall be 120°F. During Step 3 and Step 10 the equipment shall be tested in accordance with the requirements of Para 3.17.

4.14 Low Temperature Test.- The equipment shall be tested in accordance with MIL Std 810, Method 502, Procedure 1. In Step 2 the internal chamber temperature shall be -65°F for a period of 24 hours. Step 3 shall be deleted from the procedure. In Step 4 the internal chamber temperature shall be -25°F. During Step 5 and Step 7 the equipment shall be tested in accordance with the requirements of Para 3.18.

4.15 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). The EVR shall consist of the following:

a. An audit to establish that "as-built" equipment including individual parts, their use in subassemblies and assemblies (components) are in accordance with the end product drawings and the requirement of MIL-P-11268.

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b. An audit to establish that "as-built" equipment, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts and components and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government Product Assurance Organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government Verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts and components to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the equipment necessary to permit government review of the equipment, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.16 Visual and mechanical inspection.- The equipment shall be examined for the requirements of workmanship per 3.25 and the defects listed in Table VI.

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TABLE VI - Classification of Visual and Mechanical Defects

Classification	Classification
MAJOR DEFECTS	MINOR DEFECTS
1. Hub locks misaligned or do not lock.	1. Loose nuts and screws.
2. End play on axles more than 1/2"	2. Loose inserts.
3. Brake lining loose.	3. Scratches, cuts, abrasions etc. causing exposure of bare metal.
4. Broken or chipped teeth on timing belt.	4. Marking not as specified.
5. V Belts cracked chipped etc.	5. Stains on canvas cover.
6. Crank GC-4-A not secure on frame hanger.	6. Final paint film not as specified.
7. Control levers bind.	
8. Axles bind.	
9. Side of brake drum contacts side of stationary frame.	
10. Control handles do not spring back quickly to neutral position when released from driving or braking positions.	
11. Choke or throttle control on control panel binding.	
12. Positioning pin of hub locking device does not release quickly when cam handle is moved to release position.	
13. Brakes tend to engage when reel shafts are manually operated with cranks, with loaded reels in position on reel axles.	

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Classification	Classification
MAJOR DEFECTS	MINOR DEFECTS
14. Misalignment between driving and driven pulleys or gears. (Total tolerance = 1/32 inch)	
15. Misalignment of belt idler assemblies. (Center of groove on idler pulley must be in line with centers of grooves on driving and driven pulleys, within total tolerance of 1/32 inch)	
16. Timing belt too tight or too loose. (The timing belt when in operating position, shall be movable 3/8 + 1/8 inch upward from bottom flat surface of belt, with only pressure of finger of one hand).	
17. Engine position adjustment not proper for tightness or alignment of belt. (With the engine slotted base moved to maximum slack-belt position, the belt must be removable from both gears, and replaceable, by manually sliding sideways off and on the larger gear without use of any tool or prying of the belt).	
18. V belts not sufficiently slack when control arms are in neutral position. (V belts must	

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cont'd

Classification	Classification
MAJOR DEFECTS	MINOR DEFECTS
be removable first from the driven (larger) sheaves, by hand only, without the use of any tool.	
19. Slack belt retainers and guide fixtures improperly positioned for belt clearance when operating. (Should have 3/16 inch min and 5/16 inch max clearance between nearest outside surface of belt and nearest metal surface of retainers, when belts are tight on sheave, in driving position).	
20. Belt spacer guard assembly at double groove driving sheave on pillow block, improperly positioned, causing edge contact with belts when operating. Also insufficient clearance between this sheave and inside curved bottom of this belt spacer guard, to allow each belt when idle, to completely slack away from bottom contact with the sheave.	
21. Hub locking pins and reel loading guides not properly positioned to assure easy loading and unloading of reels.	
22. The 2 holes in the removable side access panel not properly positioned (centered) and with clearance for reel shaft stub extensions.	

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Classification	Classification
MAJOR DEFECTS	MINOR DEFECTS
23. Brakes not properly adjusted. 24. Pulleys cracked or loose shafts. 25. Cracked gears. 26. Chipped teeth. 27. Welds not of sufficient penetration. 28. Rough surfaces or sharp edges. 29. Fasteners missing or inoperable. 30. Covers fit improperly. 31. Stitches broken or missing. 32. Holes, tears; or breaks in canvas or webbing. 33. Bearings, frozen, binding or excessively loose. 34. Improper lubricant. 35. Improper finishing material used or finish improperly applied. 36. Brittle or mildewed spots on canvas, rope or webbing 37. Rope frayed or having broken strands in excess of the specification allowance. 38. Components of the equipment missing. 39. Corrosion, breakage, deformation, malfunctioning of any component or parts.	

4.17 Weight.- The equipment shall be weighed to determine compliance with the requirement specified in 3.10.

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4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified. (See [illegible])

5.1.1 Level A.-

5.1.1.1 Cleaning.- Reeling Machine, Cable, Engine Driven RL-207 ( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Reeling Machine, Cable, Engine Driven RL-207 ( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- Immediately after cleaning and drying, the engine shall be preserved in accordance with MIL-E-10062.

5.1.1.4 Unit packaging.- Each Reeling Machine, Cable Engine Driven RL-207( )/G shall be packaged in accordance with items 1 through 3 of table 1 and figures 1 and 2.

5.1.2 Level C.- Reeling Machine, Cable, Engine Driven RL-207()/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment handling and limited intransist storage.

5.2 Packing.- Packing shall be level A, B or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents (see 6).

5.2.1 Level A.- Reeling Machine, Cable, Engine Driven RL-207()/G shall be packed in accordance with items 4 and 5 of table 1 and figures 1 and 2. Box closure shall be as specified in the appendix of the box specification.

5.2.2 Level B.- Reeling Machine, Cable, Engine Driven RL-207( )/G shall be packed as specified in 5.2.1, except that the box in item 4 of table 1 shall be class 1.

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5.2.3 Level C.- Reeling Machine, Cable, Engine Driven RL-207 ( )/G, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.2.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use.- Reeling Machine, Cable, Engine Driven RL-207 ( )/G, is primarily intended to be mounted in a truck, 3/4 ton or larger, or it may be used on the ground to pay out or recover field wire and field cable.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendments thereto.

b. Levels of preservation and packaging and packing (See Section 5).

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: RL-207M/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Dimensional Data.- Dimensions of packaging materials in the Bill of Material Table are based on the dimensions of the equipment cited in the Table. When the dimensions of the equipment vary from those cited, the dimensions of packaging materials shall be adjusted accordingly. When running spares are specified in the Table and the quantities differ from the quantities in the bid request or contract, the quantities in the bid request shall prevail. Adjustments in material dimensions shall be made accordingly (See Section 5).

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6.5 Deviations.- Request for deviations in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed design change, to withstand the applicable tests for the selected method, including the rough handling tests in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing materials selected and economies realized (See Section 5).

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BILL OF MATERIAL
TABLE I

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Technical Literature	2	IC-1	10-1/4	8							
	Bag	2		11-1/2	9		MIL-B-117	II	C			
2	Reeling Machine, Cable, Engine Driven RL-207( )/G	1	III	40-1/2	41-3/4	42	(Bare wt. 490 lbs.) (35-1/2" D with control lever removed.)					
	Accessories											
	Belt	3										
	Screwdriver	1										
	Oilcan	1										
	Pull cord	1										
	Wrench	1										
	Pliers	1										
				Place accessories in tool box on Reeling Machine, close cover and secure.								
	Lever, Reel	1	III									
	Paperboard, Cushioning	1		18	6		PPP-P-291	I				
	Tape	2		12	2		PPP-T-76					
				Remove uppermost lever from machine as instructed in literature. Wrap in paperboard and secure to axle with tape.								
	Cover, Canvas	1		Place cover over reeling machine and secure with drawcord at base.								
3	Skid Base	1		45-3/4	44-1/2	4-1/4	PPP-B-621		2			7
				Per Figure 2.								
4	Box, Nailed Wood	1		45-3/4	44-1/2	37-1/2	PPP-B-621		2			7
	Bolts	4		5	1/2 dia		FF-B-575					
	Nuts	4			1/2 dia		FF-N-836					
	Washers	4			1/2 dia		FF-W-92					
				Secure item to box base with bolts, nuts and washers.								
5	Strapping, Steel	3		176	3/4	0.023	QQ-S-781	I	Finish A			

T - Type C - Class V - Variety G - Grade S - Style

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REELING MACHINE
RL-207( )/G

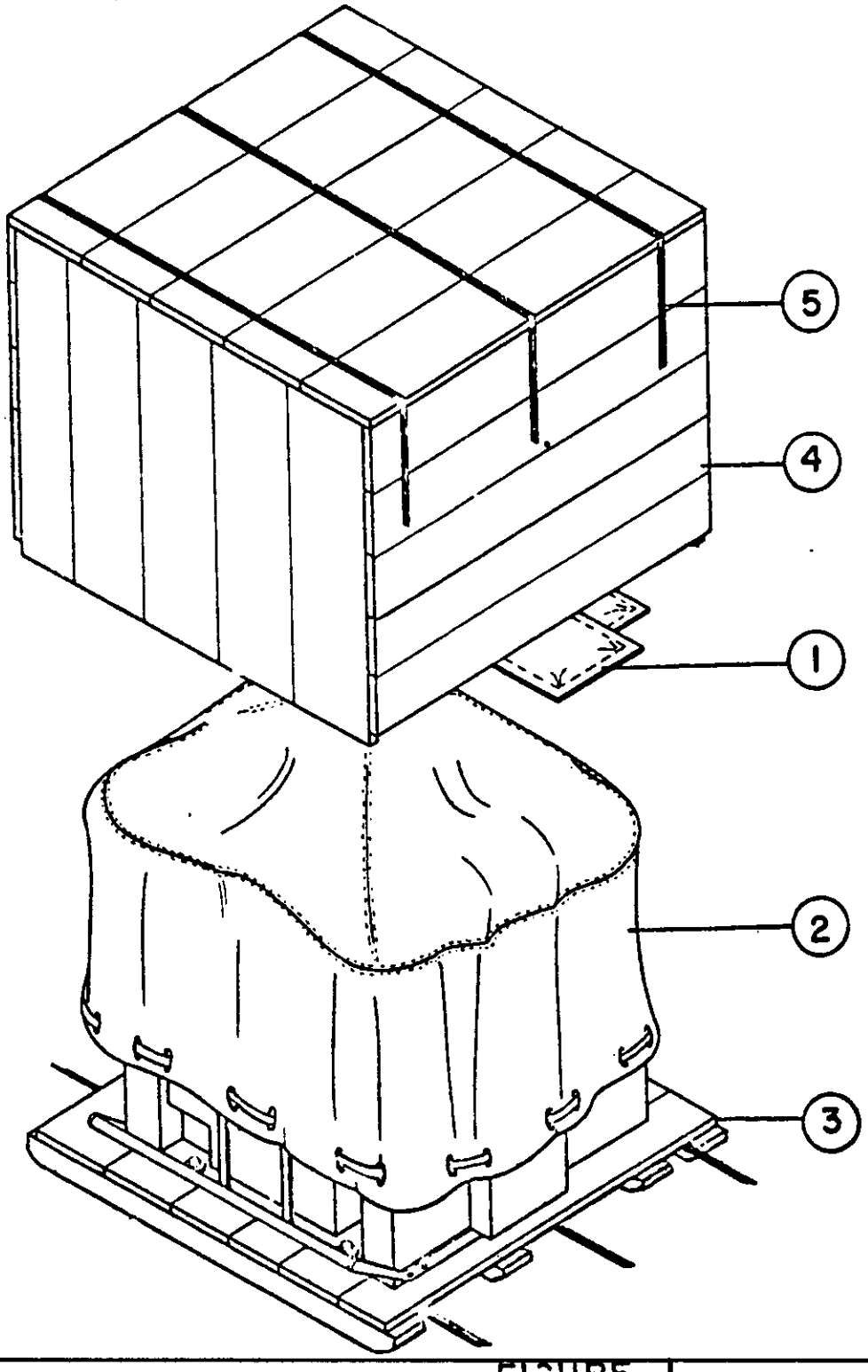


FIGURE 1

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CABLE REELING MACHINE
RL-207 ( )/G

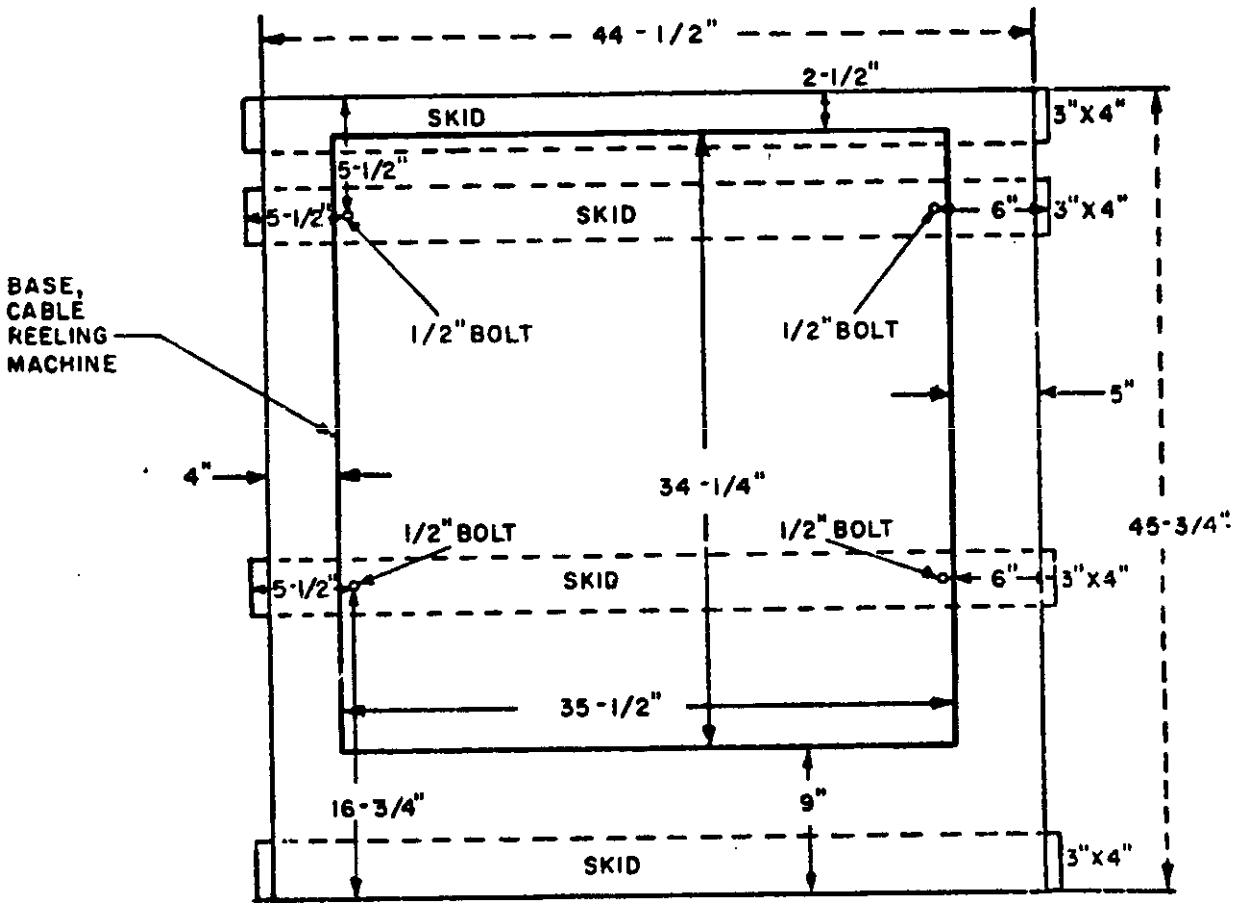


DIAGRAM OF REELING MACHINE BASE MOUNTED ON WOOD BASE, SHOWING 4 SKIDS.

FIGURE 2

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